P3↔P2 Synthesis and Series CC Final Closure

Author: Wenxin Li (Independent Researcher)
ORCID: 0009-0004-8065-3235
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Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026), and "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026).

Abstract

This note is the twentieth and final note in Series CC of the CKS derivation note corpus (#666–#685). It performs two functions. First, it summarizes the ten P3↔P2 inheritance notes (CC.09–CC.18), stating the universal pattern those notes collectively established: Paper 3's governance mechanisms are unchanged from Paper 2; Paper 3 adds inter-Self sources—specifically, Full Aspect Integration (FAI) events—to Paper 2's existing mechanisms without modifying those mechanisms. Second, it states the combined Series CC synthesis that the full twenty-note series supports: Paper 3 is Paper 1's principles applied at inter-Self scope, coordinating between Paper 2-compliant architectures. This single statement encodes the complete trilogy inheritance chain and constitutes the strongest prior-art assertion in the corpus. The note closes with a transition to Series T, which covers concept disambiguation rather than inheritance—where the same concept is scoped differently across papers rather than derived from one paper to another.

1. Position of This Note in the Series

Series CC spans twenty notes (#666–#685) and formalizes the complete inheritance chain by which Paper 3 derives from Papers 1 and 2. The first eight notes (CC.00–CC.07) covered the P3↔P1 inheritance chain—each of Paper 3's commitments inheriting directly from one or more of Paper 1's six architectural principles without redefense. Notes CC.08 through CC.18 covered the P3↔P2 inheritance chain—eleven notes establishing how Paper 3 extends each of Paper 2's major intra-Self governance commitments to the inter-Self scope.

This note, CC.19, closes the series. It does not add new inheritance edges. Its contribution is consolidation: stating the inheritance pattern that ten P3↔P2 notes collectively established, combining that pattern with the P3↔P1 pattern to produce the complete trilogy inheritance claim, and drawing the boundary that separates the inheritance work of Series CC from the disambiguation work of Series T.

2. The P3↔P2 Inheritance Notes: What CC.09–CC.18 Established

Ten notes covered how Paper 3 extends each of Paper 2’s major commitments at the inter-Self scope. The following table summarizes the territory each note formalized.

Note	Paper 2 commitment covered	Paper 3 extension at inter-Self scope
CC.09	Instinct/reasoning separation	Instinct non-crossing at the inter-Self boundary; exchange bounding to reasoning-layer content
CC.10	Three evolution mechanisms	Four-locus evolution feed adds FAI-derived content as new source to each mechanism
CC.11	Lifecycle governance	Shared substrate construction and dissolution apply the same lifecycle pattern to the shared substrate entity
CC.12	Structural hierarchy (cell, aspect, Self)	Aspect serves as the exchange unit at the inter-Self boundary, extending its role without changing its definition
CC.13	Mating primitive with three pattern variants	FAI is the inter-Self analog of mating; same three variants apply with full merge as architectural default
CC.14	Vertical evolution within a Self	FAI-origin content enters home-perimeter DNA evolution through the same vertical propagation mechanism
CC.15	Structural co-adaptation	FAI-informed adaptation uses the same DNA evolution and action-feedback mechanisms; FAI events supply additional information
CC.16	Directed selection as the DNA evolution mechanism	DNA absorption of FAI-derived content is directed selection operating with inter-Self source content

Note	Paper 2 commitment covered	Paper 3 extension at inter-Self scope
CC.17	Home perimeter and substrate source of truth	Made explicit and verifiable at inter-Self scope; shared substrate carries its own source-of-truth identity during the event
CC.18	Boundary cases at intra-Self scope	Three inter-Self governance boundaries, with governance authority as the criterion distinguishing inside from outside

No Paper 2 mechanism was modified or replaced. Every note in the P3↔P2 series had the same structural result: an existing Paper 2 mechanism now has an additional source of input, or an existing Paper 2 structural entity now has an additional scope at which it operates.

3. The P3↔P2 Universal Inheritance Pattern: Same Mechanisms, New Sources

The ten P3↔P2 notes collectively establish one pattern, stated precisely here.

Mechanisms. Paper 2's governance mechanisms operate identically at inter-Self scope as at intra-Self scope. Directed selection, action-feedback evolution, vertical evolution, the mating primitive with its three pattern variants, and lifecycle governance all function without modification when Paper 3 extends coordination across the inter-Self perimeter. No new mechanism was introduced.

Sources. Paper 3 adds inter-Self sources to Paper 2's existing mechanisms. FAI events generate content at the inter-Self boundary. That content reaches Paper 2's evolution machinery through the four-locus evolution feed: DNA-layer content from FAI feeds DNA evolution at home; action-layer content feeds action-feedback evolution at home; instinct evolution takes no FAI input by architectural commitment, per Paper 2's instinct/reasoning separation extended at the inter-Self boundary. The sources are new; the mechanisms receiving them are not.

Entities. Paper 3 applies Paper 2's governance patterns to a new entity—the shared substrate—without creating new governance patterns. The shared substrate constructs and dissolves under the same lifecycle logic that governs cells, aspects, and Selves. The six Paper 1 commitments hold within it. Paper 2's structural hierarchy (cell, aspect, Self) remains intact within each participating Self's home perimeter throughout.

The pattern phrase is: **same mechanisms, new sources.**

4. The P3↔P1 Pattern, Restated for Combination

The eight P3↔P1 notes (CC.00–CC.07) established the complement pattern. Paper 1 defines six architectural principles: human-governed, substrate-as-coordination-artifact, conflict preservation as first-class state, AI-as-substrate-mediator, tool-agnosticism, and linear-cost scaling. Paper 3's shared substrate carries all six of these commitments within scope by inheritance, without redefense. Paper 3 does not introduce new governance principles; it applies Paper 1's principles at the inter-Self scope—spanning more than one Self's home governance perimeter, with provenance crossing organizational boundaries, and with the configuration of that crossing itself under human authority.

The pattern phrase is: **same principles, new scope.**

5. The Combined Series CC Synthesis: The Trilogy Inheritance Claim

The two patterns together—same principles, new scope (P3↔P1) and same mechanisms, new sources (P3↔P2)—characterize the trilogy's complete inheritance structure. They can be combined into a single statement:

> *Paper 3 is Paper 1's principles applied at inter-Self scope, coordinating between Paper 2-compliant architectures.*

This is the trilogy inheritance claim. It says three things.

First, that Paper 3 applies Paper 1's governance principles rather than replacing them. The shared substrate carries all six Paper 1 commitments. Any deployment of Paper 3 is, at its governance layer, a Paper 1 deployment extended to span more than one Self.

Second, that Paper 3 coordinates between architectures that are Paper 2-compliant. The participating Selves are CKS-governed Selves in the Paper 2 sense—each has a three-level structure (cell, aspect, Self), DNA and action layers, the instinct/reasoning separation, lifecycle primitives, and three evolution mechanisms in productive tension. Paper 3 does not modify the internal architecture of any participating Self. FAI events operate at the boundary between Paper 2-governed Selves; what happens inside each Self's home perimeter during and after an FAI event is Paper 2 territory.

Third, that Paper 3 contributes the inter-Self context and nothing else architecturally novel. The context is the shared substrate, the FAI primitive, the three-tier conflict-handling mechanism within the shared substrate, the four-locus evolution feed at the FAI dissolution boundary, the configuration-as-substrate-content commitment with recursive applicability, and the population-scale extension. Each of these is a new entity or a new scope—not a new governance principle and not a new operational mechanism. Principles were inherited from Paper 1. Mechanisms were inherited from Paper 2. Paper 3's contribution is the architecture that applies those principles and mechanisms at the inter-Self scope.

Defensive publication force. The trilogy inheritance claim is the strongest prior-art statement in the Series CC corpus. An adversary who challenges any part of Paper 3 must address this claim, because the claim accounts for the complete derivation lineage. An adversary who claims that Paper 3's shared substrate introduces a novel governance principle must explain why the six Paper 1 commitments embedded in it do not constitute inheritance from Paper 1. An adversary who claims that FAI introduces a novel mechanism must explain why it is not the mating primitive from Paper 2 extended to the inter-Self scope with full merge as architectural default. An adversary who claims that the four-locus evolution feed is novel must explain why it is not Paper 2's three evolution mechanisms operating with FAI-derived content as additional input. The claim does not eliminate challenge; it raises the burden for every available challenge vector simultaneously.

6. What No Paper in the Trilogy Introduces

A precise characterization of what the trilogy's inheritance structure excludes is useful for defensive purposes. No paper in the trilogy introduces novel architectural principles. Each extends the prior paper or papers to a new scope or a new entity type while preserving all established commitments.

Paper 1 establishes the governance principles and the substrate-mediated coordination pattern at the single-cell scope.

Paper 2 extends those principles to a three-tier intra-Self architecture, adds the instinct/reasoning separation as a new scope-specific commitment, and introduces the lifecycle and evolution mechanisms that operate over the three-tier structure. Paper 2 does not add new governance principles to Paper 1's six; it develops them across a larger and more complex architectural object.

Paper 3 applies Paper 1's principles and Paper 2's mechanisms at the inter-Self scope. It adds the shared substrate as the architectural object, FAI as the operation over it, and the four-locus evolution feed as the connection back to home-perimeter Paper 2 machinery. It does not add new governance principles and does not modify Paper 2's mechanisms.

The trilogy is architecturally coherent and lineage-complete. Every commitment in each paper is either a fresh commitment at that paper's scope or a derivation from a commitment in a prior paper. Series CC's twenty notes have formalized that derivation chain in full.

7. Transition to Series T: From Inheritance to Disambiguation

Series CC covered inheritance. Each Series CC note answered the question: does this Paper 3 commitment derive from a specific Paper 1 or Paper 2 commitment, and if so, how? The answer in every case was yes, with the derivation mechanism stated.

Series T will cover something different. The question Series T answers is: where does the same concept appear in more than one paper, and where does its scope or meaning differ across those papers? Inheritance and disambiguation are distinct questions. A concept can be inherited by one paper from another (inheritance, Series CC territory) and also scoped differently within each paper's own domain (disambiguation, Series T territory). The distinction is between derivation lineage and scope-sensitive meaning.

Several concepts in the trilogy are used across multiple papers with scope differences that are not inheritance relationships. The aspect is both the intra-Self coordination unit (Paper 2) and the inter-Self exchange unit (Paper 3); neither use is derived from the other—both are the same object serving an additional role at a new scope. The substrate appears at cell scope (Paper 1), at Self scope as the DNA/action layer architecture's foundation (Paper 2), and at inter-Self scope as the shared substrate (Paper 3); each scope is an extension of the prior, but the substrate at each scope has scope-specific properties that need disambiguation for readers. Mating (Paper 2) and FAI (Paper 3) are analogs at different scopes; the relationship is one of structural parallel, not derivation in the strict sense. Series T will formalize these scope distinctions across approximately fifty notes, organized into concept disambiguation, substrate-mediator ladder rung transitions, trilogy-wide composition, trilogy-wide anti-patterns, and a trilogy synthesis.

Where Series CC drew the inheritance map, Series T will draw the scope map. The two maps together provide the complete cross-paper reference architecture for the CKS trilogy.

8. Operational Test for the Trilogy Inheritance Claim

A given Paper 3 commitment satisfies the trilogy inheritance claim if and only if both of the following are true:

(a) Every governance property the commitment requires is either one of Paper 1's six architectural principles held within the shared substrate by inheritance, or a direct application of those principles at the inter-Self scope.

(b) Every operational mechanism the commitment relies on is either one of Paper 2's intra-Self governance mechanisms operating at inter-Self scope without modification, or that mechanism receiving FAI-derived content as an additional source without change to the mechanism itself.

A commitment that requires a governance property not found in Paper 1's six principles, or an operational mechanism not found in Paper 2's intra-Self machinery, would fall outside the trilogy inheritance claim and would require fresh architectural defense. No such commitment has been identified in Series CC's twenty-note review.

Sources

Li, W. (April 2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.*

Li, W. (April 2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.*

Li, W. (April 2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration.*

This note is #685 in the CKS derivation note series and the final note in Series CC. It is a defensive publication of public prior art. It introduces no new architectural commitments and asserts no claims beyond the formalization of derivation relationships present in the source papers named above.